TECHNICAL DILIGENCE REVIEW - RECORD 008

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-008

1. Scope of the technical-debt review and its basis

This report is one of a series of technical reports; this one takes the technical-debt register, end-of-life dependency exposure and remediation follow-through and reports against the evidence available by 2025-12-03. This report, the present report, is the Calder Rowe Engineering Advisory workstream on the technical-debt register, end-of-life dependency exposure and remediation follow-through, delivered to Ardent Peak on 2025-12-03. Nothing in report TDR-2025-27 is presented as verified on the strength of an interview alone. R. Ostrowski has left several concerns in this report at limited evidence strength rather than dropping them or dressing them up. R. Ostrowski has stated the limitations at the point they bite rather than collecting them into a disclaimer.

This report follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. In the present report the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. Scope of the report at TDR-2025-27 is the technical-debt register, end-of-life dependency exposure and remediation follow-through, and the cut-off for evidence is 2025-12-03. Report TDR-2025-27 was prepared by Calder Rowe Engineering Advisory for Ardent Peak and covers the technical-debt register, end-of-life dependency exposure and remediation follow-through at Helioscope Robotics, Inc. as at 2025-12-03. R. Ostrowski has not smoothed the sequence of this report; it follows the order in which the evidence arrived.

Readers of the present report should note that a moderate-evidence finding may carry a high severity, and that this is not a contradiction. The evidence convention in this report is that severity describes consequence and evidence strength describes confidence, and the two are never merged. Calder Rowe Engineering Advisory will not treat an open item in report TDR-2025-27 as closed on a verbal assurance, and has said so to the engineering leadership. The reviewer has resisted the temptation to tidy report TDR-2025-27: the open items are open because the evidence is not there yet. Calder Rowe Engineering Advisory has reported the position as at 2025-12-03 and has not attempted to predict where it will be at completion.

The principal limitation on TDR-2025-27 is time: 4 weeks was enough to test the control path and not enough to test the reporting path. The reviewer notes that this report was prepared while three remediation programmes were in flight, so several positions will have moved by completion. The instruction behind report TDR-2025-27 was narrow: examine the technical-debt register, end-of-life dependency exposure and remediation follow-through and report by 2025-12-03. Ardent Peak commissioned the report at TDR-2025-27 from Calder Rowe Engineering Advisory to test the technical-debt register, end-of-life dependency exposure and remediation follow-through ahead of the Series B financing. The numbering in the report at TDR-2025-27 follows the order of the work rather than the order of importance.

The sample behind this report is 19 services and 5 sites; the selection was the reviewer's, not the company's. Report TDR-2025-27 sampled 19 services from an inventory of 34, weighted towards those on the critical control path. This report does not include a code audit; the reviewer read code where a specific claim required it and no further. Two document requests remained partly unsatisfied when report TDR-2025-27 was delivered, and the affected conclusions are marked. This report does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

Minutes of every the report at TDR-2025-27 interview were returned to the participant, and three came back with substantive corrections. Interviewees for TDR-2025-27 were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. Where report TDR-2025-27 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. The convention throughout this report is that an absence of evidence is reported as such rather than as evidence of absence. Report TDR-2025-27 was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

Schedule V - Costed technical-debt and migration register (TDR-2025-27)

Item	Area	Component	Effort estimate	Costed	Target date	Status
Partition the fleet allocator write path	scalability	Fleet Allocator	3.5 engineer-quarters	$420,000	2027-03-31	in design

Item	Area	Component	Effort estimate	Costed	Target date	Status
Replace the unmaintained scan-matching library in the localization fallback	IP and support	Localization Engine	1.5 engineer-quarters	$185,000	2026-12-31	committed
Retire the vendored geometry library in costmap fusion	support	Costmap Fusion	1 engineer-quarter	$128,000	2027-06-30	backlog
Raise robot software line coverage above sixty per cent on the perception path	quality	Costmap Fusion	2 engineer-quarters	$240,000	2027-03-31	in progress
Bring the second bench rig to the current sensor configuration	test capability	Hardware-in-the-Loop Rig	0.5 engineer-quarters	$96,000	2026-09-30	committed
Make the nightly simulation suite blocking	quality	Simulation Harness	1 engineer-quarter	$115,000	2026-11-30	blocked
Pin the full simulation build configuration and enable a determinism check	tooling	Simulation Harness	0.5 engineer-quarters	$68,000	2026-05-31	overdue
Replace the archived charting library in the operator console	security supportability	Remote Operations Console	1 engineer-quarter	$132,000	2027-01-31	backlog
Upgrade the streaming platform before its support window closes	support	Event Bus	1.5 engineer-quarters	$196,000	2027-06-30	committed
Migrate the deprecated warehouse transformation function	support	Analytics Warehouse Loader	0.5 engineer-quarters	$54,000	2026-12-31	committed
Re-converge the four forked site provisioning templates	configuration	Site Provisioning Service	0.5 engineer-quarters	$61,000	2026-09-30	in progress
Extend the edge buffer to twelve hours	resilience	Edge Gateway Agent	1 engineer-quarter	$144,000	2026-10-31	in progress
Replace the fallback scan matcher with the current implementation	performance	Localization Engine	1.5 engineer-quarters	$178,000	2026-12-31	in design
Introduce a second bench rig queue to cut firmware lead time	delivery	Hardware-in-the-Loop Rig	0.5 engineer-quarters	$88,000	2027-03-31	backlog
Add per-region credentials for the paging provider	security	Alarm Router	0.25 engineer-quarters	$34,000	2026-09-30	committed
Encrypt the on-robot localization map cache	security	Localization Engine	1 engineer-quarter	$126,000	2026-10-15	committed
Enforce a two-person rule on OTA artefact signing	security	OTA Updater	0.5 engineer-quarters	$72,000	2026-11-30	in design
Aggregate the	safety evidence	Safety Controller	0.75 engineer-	$104,000	2027-03-31	backlog

Item	Area	Component	Effort estimate	Costed	Target date	Status
safety controller fault log to the cloud continuously			quarters			
Measure closing rate for the separation monitor	safety	Speed and Separation Monitor	2 engineer-quarters	$268,000	2026-08-31	in progress
Validate and ship the wheel slip detector	safety	Wheel Odometry Filter	1.5 engineer-quarters	$162,000	2026-09-15	in progress
Refit the battery state-of-health estimator on opportunity-charge data	quality	Battery Management Service	1 engineer-quarter	$138,000	2026-09-30	in progress
Move the calibration record for the payload scale to the cloud	quality	Payload Scale Interface	0.25 engineer-quarters	$42,000	2027-01-31	backlog
Split the shared database instance used by the allocator and the charge scheduler	reliability	Charge Scheduler	0.75 engineer-quarters	$92,000	2026-12-31	committed
Publish the modified files carrying file-level copyleft obligations	IP hygiene	Costmap Fusion	0.25 engineer-quarters	$28,000	2026-09-30	committed

2. The costed register: TD-12 and the funded items

On funding, programme item TD-12 is unfunded and dependent on capital rather than engineering effort. On TD-12 the reviewer records that the site alert landed first; the buffer extension needs a storage change on the compute module. Exposure carried by programme item TD-12 is that it telemetry loss on outages beyond four hours. programme item TD-12 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Register item TD-11 is on the register because it template drift produced a go-live failure in May 2026. Funding for programme item TD-11 is not funded, and carried in the register regardless, and the register does not say so. Site Provisioning Service carries programme item TD-11, the work to re-converge the four forked site provisioning templates. Site Provisioning Service item TD-11 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

The register entry quality item TD-22 reads: move the calibration record for the payload scale to the cloud, in the quality category. In the programme Payload Scale Interface item TD-22 is ahead of two other items that depend on it. Register item TD-22 is not straightforward: a small item that keeps returning as a field support cost. programme item TD-22 illustrates why TDR-2025-27 separates the severity of a finding from the confidence behind it.

Left in place, the condition behind IP hygiene item TD-24 three modified files under a file-level copyleft licence appear unpublished. IP hygiene item TD-24 covers the work to publish the modified files carrying file-level copyleft obligations. TD-24 is funded in the current financial year in the engineering plan the reviewer was shown. TD-24 rests on a single source, and the reviewer has said so rather than implying corroboration.

TD-14 cannot start earlier than it does because it is in parallel with the release-process changes. Funded from the second quarter onward is the funding position on TD-14. Hardware-in-the-Loop Rig item TD-14 is backlog with a target date of 2027-03-31. Hardware-in-the-Loop Rig item TD-14 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-27.

The reviewer's qualification on register item TD-01 is that the load-test series identifies this as the single binding scalability constraint. TD-01 cannot start earlier than it does because it is last in a three-item chain, so its date is the least reliable on the register. Fleet Allocator carries programme item TD-01, the work to partition the fleet allocator write path.

programme item TD-01 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

What OTA Updater item TD-17 protects against is the fact that it a single key holder can sign firmware. $72,000 is the costed estimate for OTA Updater item TD-17, based on 0.5 engineer-quarters. Security item TD-17 is unfunded and dependent on capital rather than engineering effort in the engineering plan the reviewer was shown. TD-17 on OTA Updater has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Register item TD-03 cannot start earlier than it does because it is behind a platform change that has its own target date. 1 engineer-quarter is the estimate on support item TD-03, which converts to $128,000. 2027-06-30 is the target for TD-03, whose status is backlog. TD-03 was raised again with the owning engineer after the initial interview, and the position did not change.

Overdue is the status of programme item TD-07, targeted at 2026-05-31. The check is written but not enabled as at the review date is the point the reviewer would put to management on TD-07 on Simulation Harness. TD-07 cannot start earlier than it does because it is last in a three-item chain, so its date is the least reliable on the register. programme item TD-07 was inspected in the repository rather than described from the ticket, which is why the account is specific.

TD-16 on Localization Engine is partly funded, with the second phase unbudgeted; the register status and the funding position are not the same thing. Localization Engine item TD-16 is after the determinism work in the simulation harness, which is why the target date is later than the effort estimate implies. If Localization Engine item TD-16 is not done, the exposure is that it a stolen compute module discloses customer warehouse layouts. Localization Engine item TD-16 would be a candidate for the first post-close engineering review under any reasonable plan.

The reviewer's comment on IP and support item TD-02 is that the reviewer treats this as both an IP and a supportability item. Localization Engine item TD-02 carries an estimate of 1.5 engineer-quarters; the costed figure is $185,000. Funding for TD-02 on Localization Engine is funded by reallocating effort from feature work, and the register does not say so. IP and support item TD-02 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

For TD-15 the team estimates 0.25 engineer-quarters, which the reviewer converted to $34,000 for comparison. The register shows programme item TD-15 as committed, due 2026-09-30. The dependency structure places security item TD-15 last in a three-item chain, so its date is the least reliable on the register. register item TD-15 was checked against the company's own register, which had not been updated since the change landed.

Schedule T - Third-party component and licence inventory (TDR-2025-27)

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
nav-graph-solver	2.4.1	Apache-2.0	no	linked into the planner as a static library	approved 2024-02-19	Motion Planner
geom2d-fusion	0.9.7	LGPL-2.1	weak	dynamically linked by the fusion pipeline	approved 2023-08-04	Costmap Fusion
scanmatch-legacy	1.2.0	GPL-2.0	strong	linked into the localization fallback path	not approved	Localization Engine
grpc-transport	1.62.0	Apache-2.0	no	service-to-service transport across the cloud	approved 2024-05-30	Event Bus
tsdb-client	3.1.4	MIT	no	telemetry writes from the ingest service	approved 2024-09-11	Telemetry Ingest
chartkit	4.8.2	BSD-3-Clause	no	operator console visualisations	approved 2023-11-27	Remote Operations Console
physics-core-fork	0.14.3	Zlib	no	simulation physics	approved with conditions 2023-04-18	Simulation Harness
rtos-shim	1.0.9	BSD-2-Clause	no	real-time helper layer on the robot	approved 2024-01-23	Motion Planner
protobuf-runtime	25.3	BSD-3-Clause	no	message encoding across edge and	approved 2024-04-08	Config Distribution

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
				cloud		
sqlite-embedded	3.45.1	public domain dedication	no	on-robot local state	approved 2023-06-15	Edge Gateway Agent
opencv-subset	4.9.0	Apache-2.0	no	image preprocessing for grasp proposals	approved 2024-07-02	Gripper Controller
pointcloud-utils	2.1.6	MPL-2.0	weak	point cloud filtering in fusion	approved 2024-03-14	Costmap Fusion
yaml-parse	0.15.2	MIT	no	configuration parsing on the robot	approved 2023-09-29	Config Distribution
jwt-verify	5.2.1	MIT	no	token verification in the identity service	approved 2024-06-21	Identity and Authorization
kafka-client	3.6.1	Apache-2.0	no	event bus producer and consumer clients	approved 2024-02-27	Event Bus
media-transcode	6.0.0	LGPL-2.1 with an optional GPL build	weak	video snippet transcoding	approved with conditions 2024-08-16	Video Snippet Service
wms-adapter-sdk	2.2.0	commercial, per-seat	no	one customer's warehouse system adapter	approved 2023-12-05	WMS Connector
lidar-vendor-sdk	4.1.2	commercial, redistributable binary	no	lidar driver	approved 2023-05-22	Lidar Driver
safety-controller-toolchain	3.0.4	commercial, development use only	no	safety controller build toolchain	approved 2023-02-10	Safety Controller
cell-model-lib	1.8.0	Apache-2.0	no	battery state estimation	approved 2024-10-01	Battery Management Service
grid-render	3.3.5	GPL-3.0 with a commercial alternative	strong	map authoring geometry rendering	approved with conditions 2023-07-19	Map Authoring Studio
http-router	1.9.4	MIT	no	cloud service routing	approved 2024-01-09	Order Broker
otel-instrument	1.24.0	Apache-2.0	no	tracing across cloud services	approved 2024-05-14	Fleet Health Monitor
pgbouncer-managed	1.21.0	ISC	no	database connection pooling	approved 2024-03-05	Fleet Allocator
wheel-slip-model	0.3.1	AGPL-3.0	strong	prototype slip detector, not shipped	not approved	Wheel Odometry Filter
csv-export	2.0.3	MIT	no	billing and usage exports	approved 2023-10-17	Billing Usage Meter
pager-provider-sdk	3.4.0	commercial, subscription	no	alarm dispatch	approved 2023-03-28	Alarm Router
k8s-operator-framework	1.31.0	Apache-2.0	no	cloud deployment controllers	approved 2024-09-24	Site Provisioning Service
image-signing	2.5.0	Apache-2.0	no	artefact signature verification	approved 2025-01-15	OTA Updater
warehouse-sim-assets	1.1.0	CC BY-SA 4.0	strong	simulation scenario assets	approved with conditions 2024-11-06	Simulation Harness

3. End-of-life dependencies and supportability

The consequence of leaving TD-09 undone is that it an unsupported broker underneath every service. The reviewer's comment on support item TD-09 is that the upgrade requires a maintenance window at every site. TD-09 on Event Bus is the item to upgrade the streaming platform before its support window closes, classified under support. TD-09 on Event Bus rests on a single source, and the reviewer has said so rather than implying corroboration.

Funded and already staffed is the funding position on performance item TD-13. Localization Engine item TD-13 carries a 2026-12-31 target and a status of in design. The business consequence recorded against Localization Engine item

TD-13 is that it relocalisation exceeds the operator prompt timeout at sparse sites. TD-13 is the entry that changed most between the first and second document requests behind TDR-2025-27.

About TD-05 it should be said that outstanding since the April 2025 lidar timing incident. Item Hardware-in-the-Loop Rig item TD-05 in the debt register is to bring the second bench rig to the current sensor configuration, and it bears on Hardware-in-the-Loop Rig. Committed is the status of TD-05, targeted at 2026-09-30. TD-05 on Hardware-in-the-Loop Rig was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

TD-04 on Costmap Fusion covers the work to raise robot software line coverage above sixty per cent on the perception path. In the programme programme item TD-04 is behind a customer maintenance window that is not yet agreed. Left in place, the condition behind Costmap Fusion item TD-04 defect escape rate on the fusion path is the highest in the fleet. quality item TD-04 would move from a concern to a verified defect if one further measurement were supplied.

2026-12-31 is the target for TD-23, whose status is committed. Programme item TD-23 is behind a supplier release outside the company's control, which is why the target date is later than the effort estimate implies. TD-23 appears in the register and is funded from the second quarter onward. register item TD-23 was cross-checked against the engineering record supplied for TDR-2025-27, and the two agree.

The plan treats Analytics Warehouse Loader item TD-10 as unfunded and dependent on capital rather than engineering effort, which the reviewer checked against the headcount model. Programme item TD-10 is ahead of two other items that depend on it, which is why the target date is later than the effort estimate implies. The support entry at TD-10 is to migrate the deprecated warehouse transformation function. support item TD-10 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

The dependency structure places Battery Management Service item TD-21 behind a platform change that has its own target date. 1 engineer-quarter is the estimate on programme item TD-21, which converts to $138,000. Status on register item TD-21 is in progress; the target is 2026-09-30. TD-21 on Battery Management Service was tested rather than accepted, and the test is described in the working papers behind TDR-2025-27.

For TD-19 the team estimates 2 engineer-quarters, which the reviewer converted to $268,000 for comparison. Measure closing rate for the separation monitor is entered at TD-19 as an safety item. Left in place, the condition behind TD-19 on Speed and Separation Monitor the protective distance rests on a fixed approach speed assumption. safety item TD-19 rests on a single source, and the reviewer has said so rather than implying corroboration.

Sizing on register item TD-20 gives 1.5 engineer-quarters and $162,000. Wheel Odometry Filter item TD-20 is not straightforward: the prototype uses a network-copyleft component that must be replaced before shipping. The plan treats safety item TD-20 as funded in the current financial year, which the reviewer checked against the headcount model. Wheel Odometry Filter item TD-20 is the reason one section of TDR-2025-27 concludes narrowly rather than generally.

Whether quality item TD-06 happens depends on funding, and it is not funded, and carried in the register regardless. In the programme quality item TD-06 is behind a supplier release outside the company's control. The consequence of leaving programme item TD-06 undone is that it a nightly regression can reach a release candidate. quality item TD-06 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The safety evidence entry at Safety Controller item TD-18 is to aggregate the safety controller fault log to the cloud continuously. Funding for programme item TD-18 is funded only if the team's headcount plan is met, and the register does not say so. The reviewer's comment on Safety Controller item TD-18 is that the safety team supports it; it is not funded in the current plan. register item TD-18 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The estimate for programme item TD-08 is 1 engineer-quarter and $132,000. No advisory affects the current version, which is why it ranks low is the point the reviewer would put to management on register item TD-08. Reading the programme plan, programme item TD-08 is last in a three-item chain, so its date is the least reliable on the register. programme item TD-08 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Schedule K - Penetration-test findings by severity with residual risk (TDR-2025-27)

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
SEC-2025-01	High	7.4	Identity and Authorization	2025-09-16	2025-11-14	low once lifetimes are cut to one hour with refresh
SEC-2025-02	High	8.1	Release Orchestrator	2025-09-17	2025-09-19	low; redaction is enforced in the logging library
SEC-2025-03	Medium	6.2	Remote Operations Console	2025-09-18	2026-01-22	low; pinning is applied at the gateway and the agent
SEC-2025-04	Medium	5.4	Map Authoring Studio	2025-09-18	2025-11-27	medium; the shortened publication path is still outstanding
SEC-2025-05	High	7.8	Telemetry Ingest	2025-09-19	2026-02-06	low; attestation is now required on the ingest path
SEC-2025-06	Medium	6.5	Video Snippet Service	2025-09-19	2025-12-18	low; the index is scoped per tenant and the identifiers are opaque
SEC-2025-07	Medium	5.9	Site Provisioning Service	2025-09-22	2026-01-16	medium; two forked templates were not re-issued until May 2026
SEC-2025-08	Medium	6.0	Event Bus	2025-09-23	2025-10-31	low; the interface is restricted to the operations subnet
SEC-2025-09	Medium	6.3	OTA Updater	2025-09-24	pending	medium; a two-person rule is designed but not enforced in the tool
SEC-2025-10	Low	3.7	Charge Scheduler	2025-09-24	2025-10-09	low
SEC-2025-11	Low	3.1	Remote Operations Console	2025-09-25	2025-10-30	low
SEC-2025-12	Low	4.2	Analytics Warehouse Loader	2025-09-25	not planned	medium; management accepted the risk with a compensating audit trail
SEC-2026-01	High	7.1	Release Orchestrator	2026-04-14	committed 2026-08-31	high until the second-approver control lands
SEC-2026-02	High	7.6	Robot Shadow Store	2026-04-15	2026-05-29	low; authorization is checked on both the cache and durable paths
SEC-2026-03	High	8.4	Config Distribution	2026-04-15	2026-06-12	low; the flag was removed and the agent refuses unsigned bundles
SEC-2026-04	Medium	5.6	WMS Connector	2026-04-16	2026-05-20	low; payload logging requires an explicit per-tenant approval
SEC-2026-05	Medium	5.2	Alarm Router	2026-04-16	committed 2026-09-30	medium; per-region credentials are scheduled but not issued
SEC-2026-06	Medium	6.1	Simulation Harness	2026-04-17	pending	medium; the role was scoped down but still reads one production bucket
SEC-2026-07	Medium	6.7	Hardware-in-the-Loop Rig	2026-04-17	committed 2026-10-31	medium; network segmentation is the only current

Finding	Severity	Score band	Component	Discovered	Remediated	Residual risk
						control
SEC-2026-08	Medium	6.4	Identity and Authorization	2026-04-18	2026-06-30	low; role grants now require a second approver and are alerted
SEC-2026-09	Low	3.9	Fleet Health Monitor	2026-04-18	2026-05-08	low
SEC-2026-10	Low	4.0	Billing Usage Meter	2026-04-19	2026-06-05	low; object lock is applied and verified nightly
SEC-2026-11	Low	4.4	Gripper Controller	2026-04-19	committed 2026-11-30	medium; the safety controller limit is the compensating control
SEC-2026-12	Low	3.5	Edge Gateway Agent	2026-04-20	pending	medium; a wipe procedure exists but was not applied retroactively
SEC-2026-13	Low	3.3	Order Broker	2026-04-20	2026-05-15	low
SEC-2026-14	Low	3.8	Traffic Manager	2026-04-21	2026-06-18	low
SEC-2026-15	Low	3.0	Site Provisioning Service	2026-04-21	not planned	low; management judged the operational cost of enforcement higher than the risk
SEC-2026-16	Medium	5.8	Costmap Fusion	2026-04-22	2026-06-26	low; the socket binds to the loopback interface only
SEC-2026-17	High	7.9	Motion Planner	2026-04-22	2026-06-19	low; the port requires a signed local credential
SEC-2026-18	Low	3.4	Battery Management Service	2026-04-23	committed 2026-12-15	low; no fleet impact demonstrated
SEC-2026-19	Medium	5.1	Localization Engine	2026-04-23	committed 2026-10-15	medium; disk encryption is planned for the next compute revision
SEC-2026-20	Low	3.2	Payload Scale Interface	2026-04-24	not planned	low; management accepted pending the calibration service redesign

4. The services carrying the largest debt

The Integrations team carries WMS Connector on its service inventory, and Tomas Villareal answered for it in interview. WMS Connector owns its own data and exposes it only through a versioned interface, which the reviewer read. WMS Connector was examined for unsupported components; one customer integration still runs on a connector build from 2024. WMS Connector is promoted every two weeks, per tenant on request, with the ring order fixed by policy. Service-level intent for WMS Connector states that order acknowledgement returned to the warehouse system inside 1.5 s at p95. WMS Connector was one of two items where TDR-2025-27 could not obtain the underlying data within the engagement.

Underlying WMS Connector in this report are six tenant integration specifications and a month of connector error logs. WMS Connector reports 69 per cent line coverage, measured by the same instrumentation the pipeline uses. For WMS Connector the reviewer records that no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. WMS Connector depends on the customer warehouse management systems and the identity service and is depended on by the order broker and the billing usage meter. PostgreSQL 14 plus a message outbox table is the system of record for WMS Connector, a Java 21 component running on cloud, containers on a

managed Kubernetes cluster. WMS Connector carries a follow-up request that was satisfied after the draft of TDR-2025-27 was circulated.

Taking the evidence together, Telemetry Ingest no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. A time-series store with ninety-day retention and object storage beyond that is the system of record for Telemetry Ingest, a Go 1.22 with a Rust decoder component running on cloud, containers plus a stream processor. Data and Telemetry releases Telemetry Ingest weekly, with schema changes gated on a compatibility check, and no release in the sampled window bypassed the gate set. Placement of Telemetry Ingest is cloud, one ingest endpoint per region with an edge buffer on the robot, which the reviewer tested by tracing a request end to end. Reading the dependency map from Telemetry Ingest outward gives the edge gateway agent and the event bus on the input side and the fleet health monitor, the analytics warehouse loader and the alarm router on the output side. Telemetry Ingest was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Telemetry Ingest carries 19 open defects, among them a case where the edge buffer drops the oldest samples after four hours offline without raising an alert. Telemetry Ingest is owned by the Data and Telemetry team, with Farida Cisse named as the engineering owner of record. Telemetry Ingest carries a single measurable objective, namely that telemetry visible in the operator console inside 30 s of emission at p95. Interface discipline around Telemetry Ingest is adequate; the schema is versioned and change is gated on a compatibility check. Telemetry Ingest rests on the telemetry schema registry, retention configuration and a forced-offline buffer test so far as this review is concerned. Telemetry Ingest was tested rather than accepted, and the test is described in the working papers behind TDR-2025-27.

Implementation of Robot Shadow Store rests on Go 1.22; the process runs on cloud, containers with a sharded cache and persists to Redis 7 as the hot path with PostgreSQL 14 as the durable record. The topology for Robot Shadow Store is cloud, regional, sharded by site, and the reviewer confirmed it against the deployment manifests. Reading the dependency map from Robot Shadow Store outward gives the telemetry ingest path and the edge gateway agent on the input side and the fleet allocator, the traffic manager and the remote operations console on the output side. One coupling concern applies to Robot Shadow Store: Redis 7 as the hot path with PostgreSQL 14 as the durable record is shared with a second service, so a schema change is not a local decision. Marta Kellerman of Fleet Services maintains Robot Shadow Store, and the service inventory entry was current at the review date. Robot Shadow Store is the clearest instance in TDR-2025-27 of engineering practice being better than the documentation suggests.

What the reviewer had for Robot Shadow Store was the shadow consistency note, staleness measurements and a rebalance observation; the conclusions below do not reach past it. The backlog for Robot Shadow Store is 9 items deep and contains an acknowledged defect whereby a shard rebalance briefly serves stale shadow state without marking it stale. Robot Shadow Store ships weekly; the reviewer sampled the last twelve releases. Automated test coverage on Robot Shadow Store stands at 68 per cent line coverage. Read against its objective, Robot Shadow Store the component would not block the transaction, and it would appear on any competent post-close engineering plan. Robot Shadow Store was checked against the company's own register, which had not been updated since the change landed.

The Autonomy Core team carries Localization Engine on its service inventory, and Rahul Menon answered for it in interview. The evidence base for Localization Engine was the localization design note, survey reports from three sites and the reviewer's map audit. In production Localization Engine is edge, with map distribution from the cloud, unchanged since the last architecture review. Localization Engine does not share a schema with any consumer; every read by the motion planner and the fleet health monitor goes through the published interface. For Localization Engine the supportability position is an unmaintained scan-matching library is still linked in the fallback path. Localization Engine is reported here in full because a summary of it in an earlier draft of TDR-2025-27 was misread as an assurance.

Release cadence for Localization Engine is every three weeks with the planner, and the release records bear that out. The motion planner and the fleet health monitor rely on Localization Engine, which in turn relies on the lidar driver, the wheel odometry filter and the map distribution channel. Localization Engine pairs a C++17 with a Rust bridge layer implementation with on-robot Linux, pinned to a dedicated core, and its data lives in a local map cache with a periodic checkpoint. Where Localization Engine is concerned, the commitment is that pose error stays under 25 mm in mapped aisles at survey speed; the reviewer traced the metric to its source. Localization Engine shows 66 per cent line coverage in the pipeline, and the reviewer notes that the untested share includes the error handling. Localization Engine was checked against the corresponding entry in the company's own register, and the register was current.

The Map Authoring Studio codebase is TypeScript with a Python service layer throughout, deployed onto cloud, a desktop-class web application, and its durable record is PostgreSQL 14 with map revisions in object storage. Map Authoring Studio's dependency ageing was checked: a geometry rendering library one major version behind its supported line. The Applications team carries Map Authoring Studio on its service inventory, and Lucia Ferrante answered for it in interview. Map Authoring Studio was weighed on its coverage, its defect backlog and its dependency position, and there is no finding of unfitness; there is a finding that the team cannot currently demonstrate fitness. For Map Authoring Studio the stated target is that a published map revision reaches every robot at the site inside fifteen minutes, expressed against production traffic rather than a benchmark. Map Authoring Studio sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The cadence recorded for Map Authoring Studio is weekly, which matches the observed release history. Map Authoring Studio was checked for shared-database coupling, and the ownership boundary held. Map Authoring Studio sits at 49 per cent line coverage; the trend across the reviewed period was flat. Placement of Map Authoring Studio is cloud, single global deployment, which the reviewer tested by tracing a request end to end. For Map Authoring Studio the reviewer worked from the authoring workflow document, the revision history for six sites and an export audit. Map Authoring Studio is included because a reader of TDR-2025-27 should be able to disagree with the inference without repeating the work.

Simulation Harness was examined for unsupported components; the physics core depends on a library branch that stopped receiving fixes in 2023. Simulation Harness was assessed on the scenario catalogue, two weeks of nightly run records and a determinism experiment, and nothing beyond that was available in the data room. Simulation Harness commits that the nightly scenario suite completes inside four hours, and the objective is measured continuously rather than sampled. Simulation Harness shows 47 per cent line coverage in the pipeline, and that is below what this component's failure mode warrants. Release cadence for Simulation Harness is continuous, from the main branch, and the release records bear that out. Simulation Harness is the clearest instance in TDR-2025-27 of a control that exists on paper and is weaker in practice.

Inputs to Simulation Harness come from the planner and localization builds under test, and its consumers are the release orchestrator gate and the autonomy regression report. The open defect backlog for Simulation Harness holds 31 items, the most consequential being that scenario determinism breaks when the physics core is rebuilt with a different optimisation level. Simulation Harness owns its own data and exposes it only through a versioned interface, which the reviewer read. The topology for Simulation Harness is cloud batch, with a local mode for developers, and the reviewer confirmed it against the deployment manifests. Simulation Harness is owned by the Developer Productivity team, with Colin Achebe named as the engineering owner of record. Simulation Harness is the only entry in its group where the reviewer and the engineering team reached different conclusions.

The manifests place Wheel Odometry Filter edge. Robot Firmware publishes one objective for Wheel Odometry Filter: odometry drift stays under 1.5 per cent of distance travelled on a clean floor. For Wheel Odometry Filter the supportability position is none identified. For Wheel Odometry Filter the reviewer records that the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Robot Firmware tracks 8 open defects on Wheel Odometry Filter, including the acknowledged case where drift exceeds the specification on wet or dusty floors and is not flagged to the operator. Wheel Odometry Filter was checked against the corresponding entry in the company's own register, and the register was current.

Test coverage on Wheel Odometry Filter is 62 per cent line coverage and the measurement basis was confirmed with Yuki Harada. Wheel Odometry Filter pairs a C++17 implementation with on-robot Linux, in the drive control process, and its data lives in no persistent store. Inputs to Wheel Odometry Filter come from the wheel encoders and the inertial measurement unit, and its consumers are the localization engine and the motion planner. Wheel Odometry Filter publishes a stable interface, though two fields used by the localization engine and the motion planner are documented only in code comments. Wheel Odometry Filter is promoted with the firmware, twice a year, with the ring order fixed by policy. Wheel Odometry Filter was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

E-Stop Chain owns its own data and exposes it only through a versioned interface, which the reviewer read. Evidence examined for E-Stop Chain comprised the wiring diagrams, the change order file and a witnessed actuation test

on four units. No meaningful automated unit coverage, verification being by witnessed test instead is reported for E-Stop Chain, and the reviewer re-ran the measurement on a clean checkout. Reading the dependency map from E-Stop Chain outward gives the operator stop buttons and the perimeter interlocks on the input side and the safety controller and the drive contactors on the output side. Theo Bassett of Safety Engineering maintains E-Stop Chain, and the service inventory entry was current at the review date. E-Stop Chain was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Implementation of E-Stop Chain rests on VHDL with discrete relay logic; the process runs on a safety-rated hardware assembly, no general-purpose operating system and persists to no persistent store; events are latched to the safety controller. E-Stop Chain's dependency ageing was checked: none; the relay parts are current production. The published objective for E-Stop Chain is that drive power is removed within 250 ms of button actuation on every tested unit. The open defect backlog for E-Stop Chain holds 2 items, the most consequential being that one pre-production harness revision remains in the field on nine robots at two sites. E-Stop Chain sits edge, hardwired, independent of the compute module, and no alternative placement is currently supported. E-Stop Chain is the entry that changed most between the first and second document requests behind TDR-2025-27.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2025-27)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
lint and format	1 m 10 s	1 m 40 s	3.1%	0.1%	blocking	Developer Productivity
unit tests, cloud services	6 m 20 s	9 m 05 s	5.4%	1.9%	blocking	Fleet Services
unit tests, robot software	11 m 45 s	18 m 30 s	7.8%	4.6%	blocking	Autonomy Core
static analysis, robot software	14 m 05 s	22 m 10 s	6.2%	0.4%	blocking	Autonomy Core
build, cloud container images	8 m 15 s	12 m 40 s	2.7%	0.3%	blocking	Platform Infrastructure
build, robot firmware images	19 m 30 s	31 m 20 s	4.1%	0.6%	blocking	Robot Firmware
integration tests, cloud	24 m 50 s	41 m 15 s	12.6%	8.2%	blocking	Fleet Services
contract tests, warehouse connectors	9 m 40 s	15 m 55 s	9.3%	3.7%	blocking	Integrations

4. The services carrying the largest debt - part 2 of 2

This is part 2 of the 4. The services carrying the largest debt, carried over unbroken; a reader reconciling a reference should treat the parts as a single register. Nothing is restated from the earlier part.

Column definitions stated with the first part of the 4. The services carrying the largest debt govern the whole of it, and any qualification recorded there is carried forward here. Pagination alone determines where it falls.

Schedule Q - Pipeline stage duration, failure rate and gate configuration (TDR-2025-27) (continued, part 2 of 2)

Pipeline stage	Mean duration	p90 duration	Failure rate	Flaky rate	Gate	Owning team
simulation suite, short	33 m 10 s	52 m 00 s	8.9%	5.1%	blocking	Autonomy Core
simulation suite, nightly	3 h 40 m	4 h 25 m	14.2%	6.8%	advisory	Autonomy Core
hardware-in-the-loop, bench	1 h 15 m	2 h 05 m	10.4%	2.2%	blocking for firmware	Robot Firmware
dependency and licence scan	4 m 25 s	6 m 50 s	1.8%	0.2%	advisory	Security Engineering
secret scan	2 m 05 s	3 m 15 s	0.9%	0.1%	blocking	Security Engineering
container image vulnerability scan	5 m 35 s	8 m 20 s	16.7%	1.1%	advisory	Security Engineering
performance smoke, cloud	12 m 20 s	19 m 45 s	6.6%	3.4%	advisory	Platform Infrastructure
release packaging and signing	7 m 50 s	11 m 10 s	2.2%	0.2%	blocking	Developer Productivity

5. Debt with a capacity consequence

The test bed for load run LT-2026-04 carries a smaller order history than production and a synthetic warehouse endpoint, which is the main reason the reviewer does not treat it as production-equivalent. Load run LT-2026-04 was designed to change one variable, namely the order rate rather than the fleet size. What limited 1200-robot run LT-2026-04 was the allocator write path needs partitioning to go further. run LT-2026-04 has no single accountable owner, and that is the finding the reviewer would press hardest.

In the series LT-2026-05 is the run that changes the presence of a failure injection. The test bed for LT-2026-05 carries the production cloud configuration and a laboratory network profile, which is the main reason the reviewer does not treat it as production-equivalent. What limited LT-2026-05 was marginally over the p95 target under burst. LT-2026-05 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Load run LT-2026-06 sits against the 1,500 ms p95 acknowledgement objective for the warehouse connector; whether that target is the right one is a separate question. The judgement the reviewer draws from run LT-2026-06 is that the run was executed against a synthetic warehouse system rather than a customer endpoint, which flatters the tail. Scenario LT-2026-06 came out pass with observation. shard rebalance under peak load run LT-2026-06 was raised again with the owning engineer after the initial interview, and the position did not change.

Scenario LT-2026-07 is recorded as pass. Configuration for load run LT-2026-07 was 960 robots against 18,200 orders an hour in the identity service token storm case. Reading identity service token storm run LT-2026-07 for the binding constraint gives token issuance was not a limiting term at any tested scale. 960-robot run LT-2026-07 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Console session concurrency run LT-2026-08 is recorded as pass. Run LT-2026-08 is scored against the recovery objective of forty seconds for a database failover, which the company set in its own service documentation. The constraint surfaced by scenario LT-2026-08 is tested to 140 concurrent operator sessions. scenario LT-2026-08 carries a follow-up request that was satisfied after the draft of TDR-2025-27 was circulated.

Load run LT-2026-09 supports this conclusion: the result holds only for the tested order mix, which is drawn from two sites and is not representative of the newest ones. Run LT-2026-09 sits against the 1,200 ms p95 objective published for order assignment; whether that target is the right one is a separate question. The limiting term in 480-robot run LT-2026-09 was the session relay is single-writer per region. load run LT-2026-09 is one of the entries where TDR-2025-27 relies on the company's own measurement rather than an independent one.

Warehouse connector burst, one tenant is the scenario at load run LT-2026-10: 240 robots, 8,800 orders an hour, dated 2026-02-26. The result recorded for load run LT-2026-10 is fail against the 1,500 ms p95 acknowledgement target, which the reviewer traced to the underlying measurement. Warehouse connector burst, one tenant run LT-2026-10 produced a 610 ms median with 1,830 ms and 4,400 ms tails and an 0.68% error rate. load run LT-2026-10 carries a follow-up request that was satisfied after the draft of TDR-2025-27 was circulated.

A fleet simulator rather than physical robots, with the control loop unchanged describes the environment scenario LT-2026-11 ran in. Outcome of scenario LT-2026-11 was pass, and the reviewer read the raw result rather than the summary. In the series LT-2026-11 is the run that changes one platform change between otherwise identical runs. warehouse connector burst after outbox partitioning run LT-2026-11 has no single accountable owner, and that is the finding the reviewer would press hardest.

Latency in telemetry ingest at twice the current fleet run LT-2026-12 was 1.4 s to console at p50, 22 s at p95 and 48 s at p99, with an error rate of 0.08%. Set beside the other runs, telemetry ingest at twice the current fleet run LT-2026-12 varies only in whether control and telemetry shared a topic. Telemetry ingest at twice the current fleet is the scenario at scenario LT-2026-12: 2400 robots, n/a orders an hour, dated 2026-04-09. telemetry ingest at twice the current fleet run LT-2026-12 was one of two items where TDR-2025-27 could not obtain the underlying data within the engagement.

Run LT-2026-13 is not read as a headline number: the result is consistent with the incident record, which is the strongest corroboration available. Twelve sites, projected 2027 fleet run LT-2026-13 exercised the twelve sites, projected 2027 fleet case with 1440 robots and 27,000 orders an hour, on 2026-04-14. LT-2026-13 was judged against the internal objective that no telemetry is lost for an outage under four hours. 1440-robot run LT-2026-13 appears twice in the working papers for TDR-2025-27 because the first answer was corrected after a follow-up request.

The applicable target for 720-robot run LT-2026-14 is the recovery objective of forty seconds for a database failover, and the reviewer checked where that target came from. Reading four hour edge buffer drain after an outage run LT-2026-14 for the binding constraint gives drain took 71 minutes and delayed the analytics load past its window. Four hour edge buffer drain after an outage is the scenario at scenario LT-2026-14: 720 robots, 13,800 orders an hour, dated 2026-04-15. run LT-2026-14 appears twice in the working papers for TDR-2025-27 because the first answer was corrected after a follow-up request.

Scenario LT-2026-15 was judged against the recovery objective of forty seconds for a database failover. In 1000-robot run LT-2026-15 the bottleneck was inside the fifteen-minute objective at the tested fleet size. Scenario LT-2026-15 is best read against the neighbouring run in the same series, from which it differs in the warehouse system endpoint used. load run LT-2026-15 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-27.

Against the stated targets, run LT-2026-16 is pass with observation. Scenario LT-2026-16 ran against an environment with a synthetic robot fleet driven from recorded mission traces. Measurement of load run LT-2026-16 is against the recovery objective of forty seconds for a database failover rather than against a customer commitment. load run LT-2026-16 rests on a single source, and the reviewer has said so rather than implying corroboration.

Schedule P - Load-test matrix against the published latency targets (TDR-2025-27)

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
LT-2025-11	single site, steady state	120	1,850	210 ms	480 ms	760 ms	0.02%	pass
LT-2025-12	single site, peak hour	120	2,640	265 ms	690 ms	1,240 ms	0.05%	pass
LT-2025-13	single site, peak with a charger outage	120	2,640	310 ms	870 ms	1,690 ms	0.11%	pass with observation
LT-2025-14	two sites, shared region	240	4,900	290 ms	820 ms	1,510 ms	0.09%	pass
LT-2025-15	four sites, shared region	480	9,400	355 ms	1,120 ms	2,240 ms	0.18%	pass with observation
LT-2025-16	six sites, shared region	720	13,800	470 ms	1,680 ms	3,410 ms	0.34%	fail against the 1,200 ms p95 target
LT-2025-17	order burst, ten times nominal for two minutes	240	12,000	620 ms	2,350 ms	5,900 ms	1.24%	fail against the 1,200 ms p95 target
LT-2025-18	telemetry burst with control traffic on the shared topic	480	9,400	540 ms	4,100 ms	11,300 ms	0.42%	fail
LT-2025-19	region failover during peak	480	9,400	410 ms	1,950 ms	6,700 ms	2.85%	fail against the error rate target
LT-2025-20	database failover, allocator primary	240	4,900	340 ms	1,410 ms	4,200 ms	0.96%	pass with observation
LT-2026-01	six sites after the topic split	720	13,800	395 ms	1,090 ms	2,050 ms	0.14%	pass
LT-2026-02	eight sites, shared region	960	18,200	455 ms	1,340 ms	2,880 ms	0.27%	fail against the 1,200 ms p95 target
LT-2026-03	eight sites with the allocator read replicas enabled	960	18,200	400 ms	1,150 ms	2,310 ms	0.21%	pass
LT-2026-04	ten sites, shared region	1200	22,600	520 ms	1,720 ms	4,050 ms	0.48%	fail against the 1,200 ms p95 target
LT-2026-05	order burst	480	24,000	480 ms	1,240 ms	2,760 ms	0.31%	pass with

Run	Scenario	Concurrent robots	Orders per hour	p50	p95	p99	Error rate	Verdict
	after the broker queue change							observation
LT-2026-06	shard rebalance under peak load	720	13,800	430 ms	1,280 ms	3,120 ms	0.36%	pass with observation
LT-2026-07	identity service token storm	960	18,200	180 ms	520 ms	1,010 ms	0.03%	pass
LT-2026-08	console session concurrency	480	9,400	290 ms	740 ms	1,480 ms	0.07%	pass
LT-2026-09	console session concurrency at twice the plan	480	9,400	380 ms	1,460 ms	3,900 ms	0.52%	fail
LT-2026-10	warehouse connector burst, one tenant	240	8,800	610 ms	1,830 ms	4,400 ms	0.68%	fail against the 1,500 ms p95 acknowledgement target
LT-2026-11	warehouse connector burst after outbox partitioning	240	8,800	440 ms	1,210 ms	2,540 ms	0.29%	pass
LT-2026-12	telemetry ingest at twice the current fleet	2400	n/a	1.4 s to console	22 s	48 s	0.08%	pass
LT-2026-13	twelve sites, projected 2027 fleet	1440	27,000	690 ms	2,450 ms	6,100 ms	0.74%	fail
LT-2026-14	four hour edge buffer drain after an outage	720	13,800	not applicable	not applicable	not applicable	0.00%	pass with observation
LT-2026-15	map publication to a thousand robots	1000	n/a	3 m 20 s	8 m 40 s	13 m 10 s	0.10%	pass
LT-2026-16	config distribution with thirty per cent of robots offline	720	13,800	1 m 50 s	4 m 30 s	9 m 20 s	0.00%	pass with observation

6. Where debt produced an incident

Harborline Systems incident INC-2026-068 is not read as an isolated failure: management's severity rating is defensible and its remediation priority is not. Verification of the Harborline Systems incident INC-2026-068 remediation gives not landed; the detector is prototyped and unvalidated. How environmental incident INC-2026-068 surfaced matters to the assessment: robots stopped and requested assistance, which the site reported as unplanned downtime. Not a service outage; 3 h of degraded operation elapsed before environmental event INC-2026-068 was mitigated, with resolution at open at the review date. Yuki Harada committed after INC-2026-068 to add a slip detector that flags the condition and issue a site advisory on cleaning windows, due 2026-09-15. INC-2026-068 is reported here in full because a summary of it in an earlier draft of TDR-2025-27 was misread as an assurance.

In environmental event INC-2026-068, the site's cleaning schedule changed and now overlaps the peak shift, which is a process finding rather than a code finding. Wheel Odometry Filter incident INC-2026-068 affected one site and 14 robots. Wheel Odometry Filter accounts for INC-2026-068, a environmental event. Cause analysis in environmental event INC-2026-068 concluded that the filter has no wet-floor slip model and does not flag the condition to the operator. For INC-2026-068 the reviewer holds limited; the log retention window had closed before the reviewer asked for the raw data

evidence and reports the finding at that strength. Wheel Odometry Filter incident INC-2026-068 is treated as material to the investment case rather than as a routine engineering item.

INC-2025-131 traces back to a single fact: the export function did not check publication state. Duration for process bypass event INC-2025-131 splits into 3 h 22 m of impact and 44 days to close the underlying cause. For incident INC-2025-131 the reviewer holds limited; the log retention window had closed before the reviewer asked for the raw data evidence and reports the finding at that strength. Process bypass event INC-2025-131 was detected because a field engineer reported unexpected keep-out zones. Following incident INC-2025-131 the team undertook to restrict export to published revisions and shorten the publication path so the workaround is unnecessary, with Lucia Ferrante accountable by 2025-12-15. process bypass event INC-2025-131 is the entry that changed most between the first and second document requests behind TDR-2025-27.

So far as reach is concerned, Map Authoring Studio incident INC-2025-131 involved one site and 34 robots. On 2025-10-14 the record for INC-2025-131 opens: an unpublished map draft was exported and hand-loaded at a site and robots ran against unreviewed geometry. The export restriction landed on 2025-11-27; the publication work is open is the position on the Juniper Bay Apparel incident INC-2025-131 commitment as at the review date. Process bypass is the class of INC-2025-131; Map Authoring Studio is the component. Assessed on its own record, Map Authoring Studio incident INC-2025-131 shows that an operational workaround masked the defect for longer than the incident record suggests. INC-2025-131 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Crestline Fulfillment incident INC-2026-039 began on 2026-04-07 when an order cancellation raced with assignment and left an orphaned task that blocked a pick face for an hour. Underneath Crestline Fulfillment incident INC-2026-039 lies the design decision that cancellation and assignment wrote to the task record without a shared lock. Classifying incident INC-2026-039 as concurrency matters because the Order Broker backlog contains two further items of the same kind. Landed on 2026-04-26 is the position on the Crestline Fulfillment incident INC-2026-039 commitment as at the review date. Attention came to concurrency incident INC-2026-039 when the site supervisor reported a blocked location. INC-2026-039 was excluded from an earlier draft of TDR-2025-27 and reinstated after the reviewer obtained the supporting record.

The judgement recorded against INC-2026-039 is that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Concurrency incident INC-2026-039 was mitigated in 1 h 04 m; the resolution ran to 19 days. Contributing to Order Broker incident INC-2026-039, and not repaired by the remediation, is that the journal reconciliation runs nightly, so the orphan was invisible until the morning. Order Broker incident INC-2026-039 touched one site and 1 robots, and Crestline Fulfillment was the affected account. Concurrency event INC-2026-039 produced one commitment: to serialise cancellation against assignment and add an hourly orphan sweep, owned by Grace Lindqvist, dated 2026-05-31. concurrency incident INC-2026-039 is one of the entries where TDR-2025-27 relies on the company's own measurement rather than an independent one.

Policy defect incident INC-2025-063 carries a second-order finding: the charge scheduler and the allocator hold separate views of charger availability. The reviewer's reading of Fleet Allocator incident INC-2025-063 is that an operational workaround masked the defect for longer than the incident record suggests. On delivery, incident INC-2025-063 shows this: landed on 2025-06-08. The classification applied to policy defect incident INC-2025-063 is policy defect, and the affected component is Fleet Allocator. Underneath policy defect incident INC-2025-063 lies the design decision that the allocator filter excluded low-charge robots without checking whether a charge slot was actually available. incident INC-2025-063 would move from a concern to a verified defect if one further measurement were supplied.

On 2025-05-27 the record for Fleet Allocator incident INC-2025-063 opens: robots parked below the battery reserve threshold were never offered work and the site ran short of movers. Policy defect event INC-2025-063 was mitigated in 6 h 05 m; the resolution ran to 12 days. Attention came to policy defect event INC-2025-063 when an operator noticed idle robots and raised a support ticket. Remediation for policy defect event INC-2025-063 is to make the allocator consult charger availability before excluding a robot; Marta Kellerman owns it and the committed date is 2025-07-31. Grading incident INC-2025-063 on evidence rather than on consequence gives limited; the log retention window had closed before the reviewer asked for the raw data. Fleet Allocator incident INC-2025-063 is one of the entries where TDR-2025-27 relies on the company's own measurement rather than an independent one.

On delivery, incident INC-2026-051 shows this: the check landed on 2026-05-15; two templates remain forked. Site Provisioning Service incident INC-2026-051 classifies as a configuration drift event against Site Provisioning Service. Larkspur Home felt INC-2026-051, which affected 1 site or tenant with no robot taken out of service. Configuration drift event INC-2026-051 supports one conclusion and not another: the blast radius was set by a deployment topology decision made years earlier. The detection path for configuration drift event INC-2026-051 was the identity service rejected the site's first token request. Site Provisioning Service incident INC-2026-051 is included because a reader of TDR-2025-27 should be able to disagree with the inference without repeating the work.

Incident INC-2026-051 would have been shorter or smaller had it not been that template drift was not detected because the baseline check compares only the default template. Incident INC-2026-051 shows a short mitigation and a long tail: 2 h 18 m against 9 days. Configuration drift event INC-2026-051 produced one commitment: to check every template against the baseline and re-converge the four forked templates, owned by Marcus Duplessis, dated 2026-06-15. The post-mortem for configuration drift event INC-2026-051 identifies the cause as follows: the site template had been forked for a customer network requirement and drifted from the baseline. Grading configuration drift event INC-2026-051 on evidence rather than on consequence gives moderate on the cause and strong on the delivery position, which was checked in the repository. Site Provisioning Service incident INC-2026-051 is the entry that changed most between the first and second document requests behind TDR-2025-27.

Not applicable; retention only elapsed before internal incident INC-2025-152 was mitigated, with resolution at 37 days. Root cause in Video Snippet Service incident INC-2025-152 was that the lifecycle rule was applied per storage bucket and one region's bucket was created later without it. Data retention incident INC-2025-152 was raised on 2025-12-04 after one region retained operator video clips for ninety days against a thirty-day commitment. The rule landed on 2026-01-10; deletion of the historical clips is open is the position on the incident INC-2025-152 commitment as at the review date. The remediation committed after incident INC-2025-152 was to apply the lifecycle rule through the configuration baseline and delete the over-retained clips, owned by Anjali Sekhon with a due date of 2026-01-31. Video Snippet Service incident INC-2025-152 carries a follow-up request that was satisfied after the draft of TDR-2025-27 was circulated.

What INC-2025-152 tells the reviewer is that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Contributing to internal incident INC-2025-152, and not repaired by the remediation, is that bucket creation was not covered by the configuration baseline check. Data retention incident INC-2025-152 was detected because an internal privacy review found the discrepancy. Internal felt Video Snippet Service incident INC-2025-152, which affected no production site and no robot directly, the exposure being to internal systems. Data retention event INC-2025-152 classifies as a data retention event against Video Snippet Service. incident INC-2025-152 is the clearest instance in TDR-2025-27 of engineering practice being better than the documentation suggests.

On WMS Connector incident INC-2025-058 the evidence is strong, resting on the production traces the team supplied; the severity assessment is stated separately below. Classifying INC-2025-058 as data integrity matters because the WMS Connector backlog contains two further items of the same kind. Detection came through the customer reported missing picks the same afternoon, which places INC-2025-058 on the reported side of the ledger. Remediation for incident INC-2025-058 is to key suppression on the tenant and identifier pair and backfill the affected order records; Tomas Villareal owns it and the committed date is 2025-07-15. Root cause in incident INC-2025-058 was that duplicate suppression keyed on a customer-supplied identifier alone rather than on tenant and identifier. INC-2025-058 would move from a concern to a verified defect if one further measurement were supplied.

2 h 35 m elapsed before incident INC-2025-058 was mitigated, with resolution at 21 days. Data integrity incident INC-2025-058 records that on 2025-05-09, two tenants reused the same order identifier namespace and duplicate suppression discarded live orders. Scope of impact in data integrity incident INC-2025-058 was 2 site or tenant with no robot taken out of service, taken from the fleet records rather than from the narrative. The reviewer's reading of data integrity event INC-2025-058 is that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Beyond the direct cause, WMS Connector incident INC-2025-058 shows that the integration specification allowed the identifier to be tenant-scoped but did not require it. incident INC-2025-058 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Moderate on the cause and strong on the delivery position, which was checked in the repository is the evidence grade for scheduling event INC-2025-052, and the reader should not read severity into it. The two durations recorded for incident INC-2025-052 are 4 h 10 m to mitigate and 16 days to resolve. Scheduling event INC-2025-052 reached the engineering team when the site's own shift report, escalated the next morning. The judgement recorded against Charge Scheduler incident INC-2025-052 is that the customer-reported detection path is the finding, not the defect itself. Following scheduling event INC-2025-052 the team undertook to release charge windows on a shift-end event and reconcile the calendar nightly, with Nadia Semenova accountable by 2025-06-30. incident INC-2025-052 is treated as material to the investment case rather than as a routine engineering item.

Scheduling incident INC-2025-052 classifies as a scheduling event against Charge Scheduler. Underneath incident INC-2025-052 lies the design decision that the scheduler treated a shift calendar entry as immutable once the window had opened. The blast radius of Evergreen Warehousing incident INC-2025-052 was one site and 44 robots. INC-2025-052 records that on 2025-04-21, an early shift end left opportunity-charge windows reserved and the following shift started with a depleted fleet. Beyond the direct cause, scheduling event INC-2025-052 shows that the site had changed its shift pattern without notifying the deployment team. incident INC-2025-052 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

7. Debt with a security consequence

Tracking finding SEC-2025-10 to closure gives a status of remediated and a date of 2025-10-09. The reviewer's position on SEC-2025-10 against Charge Scheduler is that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. Evidence behind Low-severity finding SEC-2025-10 is that endpoint reachable and content captured, which the reviewer treats as strong. The practical consequence in finding SEC-2025-10 is that site naming and fleet size leak to anyone inside the cluster network. assessment finding SEC-2025-10 illustrates why TDR-2025-27 separates the severity of a finding from the confidence behind it.

The practical consequence in SEC-2025-12 against Analytics Warehouse Loader is that a loader defect or compromise could alter source telemetry. Weighing Analytics Warehouse Loader finding SEC-2025-12 against the compensating controls, the finding was closed within the committed window and the closure evidence was inspected. At 4.2 the score on finding SEC-2025-12 places it in the Low group. Tracking assessment finding SEC-2025-12 to closure gives a status of accepted and a date of not planned. finding SEC-2025-12 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Substantiating Medium-severity finding SEC-2025-04 is the fact that reproduced, and independently observed in a field incident, graded moderate. For SEC-2025-04 the reviewer notes that management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. Severity on finding SEC-2025-04 is Medium, scored 5.4 by the assessor. As at the review date assessment finding SEC-2025-04 was remediated and the closure entry reads 2025-11-27. Map Authoring Studio finding SEC-2025-04 was checked against the corresponding entry in the company's own register, and the register was current.

The residual position on assessment finding SEC-2025-08 is low; the interface is restricted to the operations subnet, stated after mitigation rather than assuming full success. For SEC-2025-08 against Event Bus the reviewer notes that the compensating control is real but depends on a network boundary that a single misconfiguration removes. Event Bus finding SEC-2025-08 matters because a compromised application container could reconfigure topics. Tracking SEC-2025-08 against Event Bus to closure gives a status of remediated and a date of 2025-10-31. SEC-2025-08 has no single accountable owner, and that is the finding the reviewer would press hardest.

Evidence behind Release Orchestrator finding SEC-2025-02 is that the tester extracted a working token from a retained log, which the reviewer treats as moderate. 2025-09-19 is the closure position on SEC-2025-02 against Release Orchestrator, which the ledger records as remediated. 2025-09-17 is the discovery date of Release Orchestrator finding SEC-2025-02, which finds that build logs printed the full authorization header. The reviewer's position on assessment finding SEC-2025-02 is that the underlying pattern appears in two other components that the tester did not reach. assessment finding SEC-2025-02 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

What remains after SEC-2026-17 is remediated is low; the port requires a signed local credential. Substantiating finding SEC-2026-17 is the fact that override applied on a laboratory robot and observed in plan output, graded moderate, because the demonstration was performed in a laboratory tenant rather than in production. Finding SEC-2026-17 stands remediated; 2026-06-19 is what the record gives for closure. The practical consequence in finding SEC-2026-17 is that a local compromise could alter planner limits without detection. finding SEC-2026-17 is reported at the evidence strength available and has not been promoted to fit the narrative.

2026-01-22 is the closure position on finding SEC-2025-03, which the ledger records as remediated. The reason SEC-2025-03 against Remote Operations Console was not downgraded is that an on-path attacker on a customer network could observe operator commands. At 6.2 the score on SEC-2025-03 against Remote Operations Console places it in the Medium group. SEC-2025-03 against Remote Operations Console sits against Remote Operations Console and records that remote operations session relay accepted an unpinned certificate chain. assessment finding SEC-2025-03 is the entry that changed most between the first and second document requests behind TDR-2025-27.

Remediated is the status of SEC-2026-13 against Order Broker, dated 2026-05-15. The reason SEC-2026-13 was not downgraded is that a large payload degrades broker throughput. Order Broker finding SEC-2026-13 leaves residual risk that the reviewer assesses as low. On 2026-04-20 the tester recorded Low-severity finding SEC-2026-13: order broker accepted an oversized payload without a limit. Low-severity finding SEC-2026-13 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Remediation status on finding SEC-2026-11 is open; the record shows committed 2026-11-30. On SEC-2026-11 against Gripper Controller the conclusion is that the risk acceptance lacks an expiry date, which is a process finding in its own right. Residual exposure on SEC-2026-11 reads medium; the safety controller limit is the compensating control. The reason SEC-2026-11 was not downgraded is that a non-privileged process could alter a force limit parameter. SEC-2026-11 against Gripper Controller was inspected in the repository rather than described from the ticket, which is why the account is specific.

Evidence behind finding SEC-2026-08 is that role grant performed in a staging tenant, which the reviewer treats as strong. Weighing finding SEC-2026-08 against the compensating controls, the item is open past its committed date and no revised date has been agreed. What remains after assessment finding SEC-2026-08 is remediated is low; role grants now require a second approver and are alerted. Rating for SEC-2026-08 is Medium, band 6.4; the reviewer re-read the scoring rationale and accepted it. Identity and Authorization finding SEC-2026-08 is the clearest instance in TDR-2025-27 of engineering practice being better than the documentation suggests.

Hardware-in-the-loop rig controller runs an unsupported operating system is the substance of Medium-severity finding SEC-2026-07, raised on 2026-04-17. The practical consequence in Medium-severity finding SEC-2026-07 is that an unpatched laboratory host sits on the engineering network. Remediation status on assessment finding SEC-2026-07 is open; the record shows committed 2026-10-31. SEC-2026-07 is not fully retired by its remediation: medium; network segmentation is the only current control. Medium-severity finding SEC-2026-07 would be a candidate for the first post-close engineering review under any reasonable plan.

Medium is the rating SEC-2025-07 carries, at 5.9 on the assessor's own scale. SEC-2025-07 leaves residual risk that the reviewer assesses as medium; two forked templates were not re-issued until May 2026. Assessment finding SEC-2025-07 stands remediated; 2026-01-16 is what the record gives for closure. Site Provisioning Service finding SEC-2025-07 was opened on 2025-09-22 and states that site provisioning stored a shared bootstrap secret in the template. SEC-2025-07 against Site Provisioning Service is the reason one section of TDR-2025-27 concludes narrowly rather than generally.

SEC-2025-09 is assessed as follows: the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. SEC-2025-09 is partially remediated, with a closure date of pending. Rating for finding SEC-2025-09 is Medium, band 6.3; the reviewer re-read the scoring rationale and accepted it. A single key holder could sign an unreviewed firmware artefact is the exposure Medium-severity finding SEC-2025-09 describes. Medium-severity finding SEC-2025-09 is the reason one section of TDR-2025-27 concludes narrowly rather than generally.

8. Debt with a safety consequence

Field history for HZ-08 is that it has produced no event, though the control has activated repeatedly. Register entry HZ-08 is assessed at Equipment damage severity with Frequent exposure and High controllability. After mitigation, HZ-08 is Tolerable; the register status is verified. Frequent-exposure hazard HZ-08 was excluded from an earlier draft of TDR-2025-27 and reinstated after the reviewer obtained the supporting record.

Serious injury hazard HZ-13 is the entry for the case that robot ignores a temporary barrier placed by site staff. Protection at HZ-13 comes from detection of unmapped obstacles with a stop, plus a site procedure for barriers. In the reviewed period Serious injury hazard HZ-13 has produced no event, though the control has activated repeatedly. hazard HZ-13 was raised again with the owning engineer after the initial interview, and the position did not change.

Hazard HZ-02 in the register is not clean: the reviewer confirmed mapping at four of six sites and could not evidence the remaining two. Occasional-exposure hazard HZ-02 depends for its mitigation on aisle-end speed reduction plus a mirrored sensing field at mapped blind corners. Serious injury hazard HZ-02 identifies the hazard that robot contacts a pedestrian emerging from a blind cross-aisle. register entry HZ-02 is the reason one section of TDR-2025-27 concludes narrowly rather than generally.

Where register entry HZ-15 bites is every site with a mezzanine or ramp. HZ-15 depends for its mitigation on egress routes marked as keep-out and a stop-and-clear behaviour on a fire signal. The register places at hazard HZ-15 in the register the scenario in which robot obstructs an emergency egress route. Remote-exposure hazard HZ-15 would move from a concern to a verified defect if one further measurement were supplied.

Against HZ-16 the team relies on atomic update with rollback and a version consistency check before mission acceptance. Hazard HZ-16 is assessed at Equipment damage severity with Remote exposure and Low controllability. The six sites with narrow-aisle geometry is the exposed population for Equipment damage hazard HZ-16. register entry HZ-16 was checked against the company's own register, which had not been updated since the change landed.

Method of verification on hazard HZ-09: ramp testing on a wet and a dry surface at two sites. Serious injury harm, Remote exposure and Medium controllability are the parameters recorded for register entry HZ-09. Has produced no event, though the control has activated repeatedly is the incident history behind register entry HZ-09. hazard HZ-09 in the register is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The population exposed to Occasional-exposure hazard HZ-12 is every robot variant currently in production. HZ-12 depends for its mitigation on reverse motion audible warning plus a rear sensing field. Evidence of verification at hazard HZ-12 consists of witnessed reverse test plus a review of the dock layout at four sites. HZ-12 is the reason one section of TDR-2025-27 concludes narrowly rather than generally.

Entry HZ-22 in the hazard register reads: site network loss leaves robots operating without fleet coordination. One qualification on hazard HZ-22 the reviewer would not remove is that the tested duration is shorter than the five-hour outage seen in April 2026. Has never activated in the field and is evidenced only on the bench is the incident history behind register entry HZ-22. register entry HZ-22 is one of the entries where TDR-2025-27 relies on the company's own measurement rather than an independent one.

Fatality hazard HZ-05 was scoped to the manipulator-equipped variant only, and the reviewer checked the scoping against the fleet record. The reviewer's note on register entry HZ-05 is that nine field units retain a pre-production harness revision and are tracked to retrofit. The design measure recorded at hazard HZ-05 in the register is hardwired dual-channel stop chain independent of the compute module. hazard HZ-05 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-27.

Protection at Serious injury hazard HZ-18 comes from protective distance calculation with a stated approach speed assumption. The three risk terms on Occasional-exposure hazard HZ-18 are severity Serious injury, exposure Occasional and controllability Low. Has never activated in the field and is evidenced only on the bench is the incident history behind Serious injury hazard HZ-18. Occasional-exposure hazard HZ-18 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Serious injury hazard HZ-06 produced a single event before the current control was fitted, and the register was not updated to reflect it. Method of verification on Serious injury hazard HZ-06: procedure walk-through with three service

technicians and a lockout inventory. Exposure under hazard HZ-06 extends to the six sites with narrow-aisle geometry. register entry HZ-06 is the entry that changed most between the first and second document requests behind TDR-2025-27.

Evidence of verification at Occasional-exposure hazard HZ-07 consists of simulation coverage of boundary cases plus a site survey audit. Against Occasional-exposure hazard HZ-07 the team relies on map keep-out enforcement plus a boundary confidence check in localization. Equipment damage hazard HZ-07 is the entry for the case that robot travels into an unmapped area and contacts fixed plant. hazard HZ-07 was raised again with the owning engineer after the initial interview, and the position did not change.

9. Debt exposure by site

2025-01 is the go-live month at Tracy, California, where Harborline Systems now runs 58 robots. Capacity headroom at Tracy, California is 36%, and the warehouse system is vendor B. Tracy, California sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

64 robots are deployed at Lehigh Valley, Pennsylvania, which went live in 2024-02. Against its peak profile, Lehigh Valley, Pennsylvania retains 38% headroom on a vendor B integration. Lehigh Valley, Pennsylvania was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Against its peak profile, Chino, California retains 54% headroom on a vendor D integration. At Chino, California the fleet is 33 robots, and the site has been live since 2026-01. Chino, California was inspected in the repository rather than described from the ticket, which is why the account is specific.

What distinguishes Kenosha, Wisconsin is that it is the second opportunity-charging site. Dunbar Industrial operates 41 Helioscope robots at Kenosha, Wisconsin, commissioned in 2024-11. Kenosha, Wisconsin was checked against the company's own register, which had not been updated since the change landed.

On Whitestown, Indiana the record shows that it is the site with the standing map publication workaround. At Whitestown, Indiana the fleet is 44 robots, and the site has been live since 2025-09. Whitestown, Indiana has no single accountable owner, and that is the finding the reviewer would press hardest.

Remaining capacity at Reno, Nevada is 57%, measured against the site's peak order profile. Site ST-11 is Reno, Nevada: Blue Mesa Components, 29 robots, live from 2025-05. Reno, Nevada was tested rather than accepted, and the test is described in the working papers behind TDR-2025-27.

Headroom at Groveport, Ohio stands at 44%; vendor A is the integrated warehouse system. 52 robots are deployed at Groveport, Ohio, which went live in 2024-09. Groveport, Ohio has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Newnan, Georgia is narrow-aisle geometry, and the source of the August 2025 mutual-yield stall, which is why the reviewer sampled it. Newnan, Georgia runs 66 robots for Ironwood Grocery Group and opened in 2025-07. Newnan, Georgia was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

What distinguishes Mesquite, Texas is that it is in acceptance testing at the review date. Against its peak profile, Mesquite, Texas retains 71% headroom on a vendor C integration. Mesquite, Texas would move from a concern to a verified defect if one further measurement were supplied.

The Wilmington, Ohio site integrates with vendor B and shows 14% headroom. 112 robots are deployed at Wilmington, Ohio, which went live in 2023-08. Wilmington, Ohio was raised again with the owning engineer after the initial interview, and the position did not change.

68% of headroom remains at Lockbourne, Ohio, whose connector targets vendor A. Granite Peak Distribution operates 19 Helioscope robots at Lockbourne, Ohio, commissioned in 2026-04. Lockbourne, Ohio is included because a reader of TDR-2025-27 should be able to disagree with the inference without repeating the work.

Shepherdsville, Kentucky was included in the sample because it is manipulator-equipped robots are concentrated here. Remaining capacity at Shepherdsville, Kentucky is 41%, measured against the site's peak order profile. Shepherdsville, Kentucky was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

For Ontario, California the material fact is that it is the site whose shift pattern change caused the April 2025 charge incident. The Ontario, California site integrates with vendor C and shows 26% headroom. Ontario, California was excluded from an earlier draft of TDR-2025-27 and reinstated after the reviewer obtained the supporting record.

Vendor A is the warehouse system at Meridian, Idaho, where headroom is 22%. At Meridian, Idaho the fleet is 96 robots, and the site has been live since 2022-09. Meridian, Idaho is reported here in full because a summary of it in an earlier draft of TDR-2025-27 was misread as an assurance.

10. Items the company has chosen not to address

The reviewer asked in each case whether a declined item had a compensating control, and TDR-2025-27 reports the answer even where it is unsatisfactory. Where management has declined to act, the present report preserves the decision and the stated rationale so a post-close plan can revisit it. R. Ostrowski notes in TDR-2025-27 that the company's own reporting is adequate for three of the four recommended indicators. Calder Rowe Engineering Advisory recommends that the 4 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. R. Ostrowski has kept the working papers behind the present report in a form that a second reviewer could re-perform.

Three of the open items in the report at TDR-2025-27 would close on a single document that the company has agreed to produce. Where report TDR-2025-27 cannot conclude, it says what evidence would let it conclude. The present report recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. The pack report TDR-2025-27 proposes is deliberately short, because a longer one would be produced and not read. Calder Rowe Engineering Advisory raised each finding with the owning engineer before it was written up, and the responses are recorded.

The report at TDR-2025-27 reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. In this report the reviewer records the basis of each conclusion inline, so a reader can disagree with the inference without re-doing the work. R. Ostrowski chose the sample for TDR-2025-27 after reading the dependency map, so the selection is not proportionate to headcount or to code volume. The population for TDR-2025-27 was the service inventory as at 2025-12-03; 19 entries were examined and the rest were reviewed only at inventory level. The evidence behind the present report is retained by Calder Rowe Engineering Advisory and is available to Ardent Peak on request.

Reporting under TDR-2025-27 should distinguish a target that was met from a target that was lowered, which the current pack does not. Monitoring recommended by the present report relies on metrics the company already emits, so no new instrumentation is required to start. Grading in the report at TDR-2025-27 is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. R. Ostrowski treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and report TDR-2025-27 keeps the two apart. The technical-debt register, end-of-life dependency exposure and remediation follow-through is reported here at the level of detail Ardent Peak asked for and no deeper.

11. Conclusions on the debt position

The present report concludes that the estate is materially better than it was twelve months ago and the improvement has not yet reached the test and release infrastructure. R. Ostrowski would summarise the present report in one sentence: the estate is materially better than it was twelve months ago and the improvement has not yet reached the test and release infrastructure. Interviews for TDR-2025-27 were conducted with the owning engineer present rather than with a manager speaking on their behalf. Report TDR-2025-27 draws on 22 interviews, including the engineering owners of every component examined in detail. Where the present report uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.

The costed remediation implied by report TDR-2025-27 is $880,000 over the coming four quarters, which is within the ordinary engineering budget. For the transaction, this report identifies 3 matters the reviewer would raise before closing and 4 for the post-close engineering plan. Management's declinatures are reported in report TDR-2025-27 at face value; the reviewer's disagreement is stated separately and does not overwrite them. TDR-2025-27 records 3 items the

engineering team declined to remediate, each with the reason given and the reviewer's view of it. Nothing in the present report is intended as a legal opinion or as advice on the adequacy of insurance cover.

4 indicators are recommended in this report; the reviewer deliberately avoided proposing anything that needs a new pipeline. TDR-2025-27 would have the monitoring pack include declined items with their compensating controls, so that acceptance decisions age visibly. TDR-2025-27 did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. Work on the present report ran over 4 weeks and included 22 interviews and read-only access to the engineering repositories. The technical-debt register, end-of-life dependency exposure and remediation follow-through was examined over 4 weeks, which sets a real limit on the depth reached in the later sections.